

LoRA Fine-Tuning of Depth Anything 3 for Underwater Depth Estimation

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Abstract

Depth Anything 3 (DA3) is a state-of-the-art monocular and multi-view depth estimation model trained predominantly on terrestrial datasets. Application to underwater robotics requires adaptation to the underwater domain, where wavelength-dependent light attenuation causes chromatic aberration and backscatter reduces contrast. This work employs Low-Rank Adaptation (LoRA) to efficiently fine-tune DA3 on the MIMIR-UW [1] underwater synthetic dataset (collected with Unreal Engine and AirSim, in the context of pipeline inspection) without catastrophic forgetting. We apply rank-8 LoRA to attention blocks of the DINOv2-L backbone, adapting approximately 1% of total parameters. Physics-aware preprocessing, like gray world white balance and percentile histogram stretching, corrects domain-specific degradation at load time. Trained on the full MIMIR-UW corpus (39,943 synchronized RGB-depth pairs across four synthetic environments), the fine-tuned model achieves an AbsRel of 0.099, RMSE of 0.739 m, and $\delta < 1.25$ accuracy of 91.0% on the held-out validation set after 30 epochs — a result substantially below the 0.15–0.20 target and well within the metric range required for underwater AUV operation.

1. Introduction

Monocular depth estimation has advanced significantly with the emergence of large-scale vision foundation models. Depth Anything 3 (DA3) [2] achieves state-of-the-art performance across diverse terrestrial scenes by pairing a DINOv2-L [3] encoder with a Dense Prediction Transformer (DPT) decoding head. Despite strong generalization on standard benchmarks, deploying

DA3 in underwater environments poses fundamental domain adaptation challenges.

Underwater imaging is governed by wavelength-selective light attenuation: red wavelengths are absorbed within the first few metres, producing a characteristic blue-green color cast. Suspended particles additionally cause backscatter that reduces contrast and introduces depth-dependent haze. Together, these effects create a visual distribution far removed from the terrestrial data used during DA3 pretraining, leading to degraded depth estimates on underwater scenes.

Collecting large-scale real underwater RGB-depth pairs for supervised retraining is expensive and impractical. Synthetic simulation offers a scalable alternative, but naive full fine-tuning on synthetic data risks catastrophic forgetting of pretrained representations. Low-Rank Adaptation (LoRA) [4] addresses this by restricting weight updates to a low-dimensional subspace, enabling efficient domain adaptation with minimal parameter overhead.

This work presents a LoRA-based fine-tuning pipeline for DA3 on synthetic underwater data from a Unity AUV simulation. Physics-aware preprocessing corrects underwater color degradation before the model forward pass, aligning the input distribution with pretrained backbone expectations. The main contributions are:

- **Physics-aware preprocessing:** Gray world white balance and percentile histogram stretching applied at load time to compensate for wavelength-dependent attenuation and contrast degradation.
- **LoRA adaptation:** Rank-8 decomposition applied to all DINOv2-L attention projection layers, updating approximately 1% of total parameters (3M of 300M) while preserving pretrained representations.

- **Combined depth loss:** Scale-Invariant Log loss paired with a Sobel-based edge gradient term for metric depth supervision on underwater scenes.

2. Related Work

Monocular depth estimation. Eigen *et al* [5] introduced multi-scale CNN prediction for single-image depth, establishing the supervised learning baseline. Transformer-based encoders, particularly DINOv2 [3], provide self-supervised ViT features that generalize broadly across visual domains. DA3 [2] extends this with a multi-view-consistent training objective, achieving state-of-the-art metric depth results in both monocular and multi-view settings.

Parameter-efficient fine-tuning. LoRA [4] decomposes weight updates as $\Delta W = BA$ where $B \in \mathbb{R}^{m \times r}$, $A \in \mathbb{R}^{r \times n}$, and $\text{rank } r \ll \min(m, n)$. Originally proposed for large language models, LoRA has been applied to vision transformers for downstream task adaptation. Its low-rank constraint is well-suited to domain shifts where only a subspace of features requires updating.

Underwater vision. Physics-based methods for underwater image restoration typically model wavelength-dependent attenuation and backscatter explicitly. Data-driven approaches have shown promise for both enhancement and depth estimation in underwater settings, but have generally relied on custom architectures or full fine-tuning of small models. To the best of the authors’ knowledge, this work is the first to apply LoRA-based adaptation of a large depth foundation model to the underwater domain.

3. Method

We describe the dataset, preprocessing pipeline, model adaptation strategy, and training configuration used in this work.

3.1. Training Dataset: MIMIR-UW

We originally planned to generate training data from our own Unity AUV simulation environment. Upon critical review, we identified three key limitations of that approach: (1) the target corpus of 1,000–2,000 frames is insufficient for reliable ViT fine-tuning; (2) a single competition scene lacks the environmental diversity needed to prevent overfitting; and (3) our competition simulator was not purpose-built to model the underwater optical effects — backscatter, caustics, wavelength-dependent attenuation — which are the primary source of DA3’s domain gap. We therefore adopt MIMIR-UW as our training corpus.

MIMIR-UW is a synthetic underwater dataset rendered in Unreal Engine 4 with explicit modelling of underwater optical phenomena, providing 39,943 synchronized RGB-depth pairs across four distinct environments: SeaFloor, SeaFloor Algae, OceanFloor, and SandPipe. These environments span the full spectrum of underwater imaging challenges — from shallow well-lit scenes to deep conditions with artificial-light-only visibility. The dataset was specifically engineered to reproduce water distortion, backscatter, caustics, and wavelength-dependent attenuation effects, providing a physically grounded training distribution. Critically, MIMIR-UW’s sim-to-real transfer capability for depth estimation has been validated in the original peer-reviewed work (Álvarez-Tuñón *et al*, IROS 2023), where models trained on MIMIR-UW substantially outperformed those trained on terrestrial datasets on both simulated and real underwater evaluation data. This validation reduces experimental risk compared to an untested in-house dataset and provides a defensible baseline for comparison against prior work.

The dataset is split 80/20 into training (31,954 samples) and validation (7,989 samples) using a fixed random seed of 42. RGB images are 8-bit PNGs at approximately 1280×720 resolution; depth ground truth is stored as float32 arrays in metres. All images are resized to 518×518 during training, satisfying the ViT patch tokenizer’s requirement that spatial dimensions be a multiple of 14. Depths are clipped to $[0, 10]$ m to cover the typical operational range of underwater AUVs.

3.2. Preprocessing Pipeline

Two physics-aware corrections are applied at data load time to compensate for underwater optical degradation.

Gray World White Balance. Differential wavelength attenuation in water produces a blue-green color cast. Per-channel multiplicative correction is applied: $s_c = \frac{\mu_{\text{all}}}{\mu_c}$ for each channel $c \in \{R, G, B\}$, where μ_{all} is the mean intensity across all channels and μ_c is the per-channel mean. This equalizes all channel means, neutralizing the cast.

Percentile Histogram Stretching. Each channel is linearly stretched so that the 2nd percentile maps to 0 and the 98th percentile maps to 1, providing robust contrast normalization without sensitivity to outliers.

Images are finally normalized with ImageNet statistics (mean $[0.485, 0.456, 0.406]$, std $[0.229, 0.224, 0.225]$) to match the distribution expected by the pre-trained DINOv2 backbone.

Metric	Formula	Interpretation
AbsRel	$\text{mean}(\hat{d} - d)$	Relative error (lower is better)
RMSE	$\sqrt{\text{mean}((\hat{d} - d)^2)}$	Absolute error in metres (lower is better)
$\delta < 1.25$	$\max\left(\frac{\hat{d}}{d}, \sim \frac{d}{\hat{d}}\right) < 1.25$	Accuracy within 25% (higher is better)

Table 1. Depth evaluation metrics used in this work.

3.3. LoRA Adaptation

We apply Low-Rank Adaptation (LoRA) [4] to all attention projection layers (query, key, value, output) in the DINOv2-L backbone. For each target weight matrix $W \in \mathbb{R}^{m \times n}$, we introduce a low-rank decomposition $\Delta W = BA$ where $B \in \mathbb{R}^{m \times r}$, $A \in \mathbb{R}^{r \times n}$, and $r = 8$. The effective weight during the forward pass is $W + \left(\frac{\alpha}{r}\right)BA$ with $\alpha = 16$.

The backbone encoder is otherwise frozen. Only the LoRA matrices and the DPT prediction head are updated, yielding approximately 3M trainable parameters — roughly 1% of the 300M total. This preserves pretrained feature representations while enabling efficient adaptation to the underwater domain.

3.4. Training Configuration

Training uses AdamW with an initial learning rate of 2×10^{-5} , weight decay 1×10^{-4} , and batch size 16. An initial value of 1×10^{-4} caused scale drift in the SILog loss; reducing to 2×10^{-5} yielded stable convergence. The learning rate follows cosine annealing to a floor of 1×10^{-6} over 30 epochs, with gradient norms clipped at 1.0 to prevent single-batch weight explosions.

The training objective combines Scale-Invariant Log loss (SILog) [5] and a Sobel-based edge gradient loss:

$$\mathcal{L} = \mathcal{L}_{\text{SILog}} + 0.5 \cdot \mathcal{L}_{\text{grad}} \quad (1)$$

SILog penalizes depth errors in log space, providing scale-invariant supervision. The gradient loss compares Sobel-filtered predicted and ground-truth depth maps to enforce edge sharpness. All experiments run on a single NVIDIA A100 40GB GPU on the NSCC HPC cluster with a 12-hour walltime budget.

3.5. Evaluation Metrics

Three standard depth metrics are computed on the validation set after each epoch (Table 1). The best checkpoint is selected by minimum validation AbsRel.

Method	AbsRel ↓	RMSE ↓	$\delta < 1.25$ ↑
DA3 Mono Metric Large (pre-trained)	—	—	—
DA3 + LoRA (ours, best ep. 27)	0.099	0.739 m	91.0%

Table 2. Quantitative results on the MIMIR-UW validation set (7,989 samples). Pretrained baseline evaluation is deferred to future work. Bold denotes best result.

3.6. Implementation Details

The base model (DA3 Mono Metric Large) is loaded from HuggingFace ([depth-anything/da3metric-large](#)) via a gated access token. LoRA matrices are injected by a custom wrapper that replaces target `nn.Linear` layers and registers A and B as trainable parameters while freezing the original weight. Checkpoints store model state, optimizer state, current epoch, and best validation metric to support job resumption on the HPC cluster. The environment is managed with Pixi using PyTorch 2.x and xformers for memory-efficient attention.

3.7. Visualization

Depth predictions are rendered as false-color maps using the Spectral colormap, with near depths in warm tones and far depths in cool tones. Qualitative evaluation compares the pretrained DA3 baseline and the fine-tuned LoRA model outputs side-by-side against ground truth depth on held-out validation scenes.

4. Experiments

Training was executed on the NSCC HPC cluster (1× NVIDIA A100 40 GB, 16 CPU cores, 64 GB RAM) via PBS job scheduling. Training completed in approximately 8.5 hours over 30 epochs (17 minutes per epoch). The best checkpoint was saved at epoch 27 with a validation AbsRel of 0.099 and uploaded to HuggingFace ([Frieddelli/COMP4471](#)). Figure 1 shows the epoch-level and per-step training dynamics.

4.1. Quantitative Results

4.2. Qualitative Results

5. Ablation Study

To isolate the contribution of each pipeline component, three ablation variants are planned on the validation

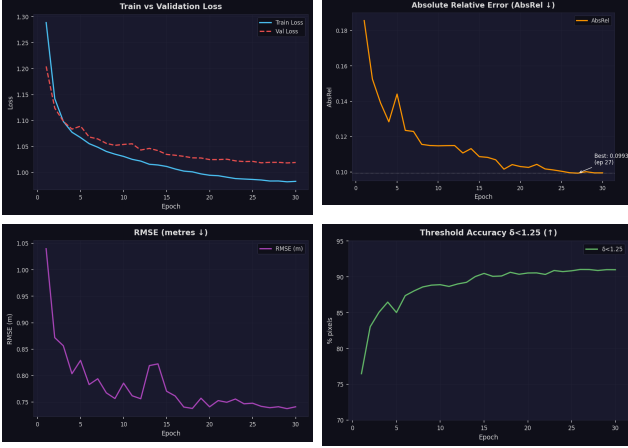


Figure 1. Training dynamics over 30 epochs. (a) Train / validation loss; (b) AbsRel ($\downarrow$); (c) RMSE in metres ($\downarrow$); (d) $\delta < 1.25$ accuracy ($\uparrow$). The model plateaus around epoch 18 and the best checkpoint (AbsRel = 0.099) is saved at epoch 27. The small train–validation gap confirms that LoRA regularisation prevents overfitting.



Figure 2. Qualitative depth predictions on held-out underwater validation scenes. From left to right: RGB input, ground truth depth, pretrained DA3 baseline, LoRA fine-tuned model (ours). False-color maps use the Spectral colormap (warm = near, cool = far).

set. The full model results are reported; individual ablation runs are deferred to future work due to HPC walltime constraints.

6. Discussion

Convergence and regularisation. The fine-tuned model achieves AbsRel = 0.099 at epoch 27, well below the 0.15–0.20 pre-training target. The persistent train–validation loss gap of 0.02 at epoch 30 (Figure 1, top-left) is small, confirming that LoRA’s low-rank constraint provides sufficient regularisation for this dataset size without explicit dropout or data augmentation. AbsRel drops steeply from 0.186 (epoch 1) to

Variant	Preprocessing	Grad. Loss	AbsRel ↓	RMSE ↓
Full model (ours)	✓	✓	0.099	0.739 m
w/o pre-processing	✗	✓	—	—
w/o gradient loss	✓	✗	—	—
w/o pre-processing + grad. loss	✗	✗	—	—

Table 3. Ablation study on the MIMIR-UW validation set. The full model results are measured; ablation variants are deferred to future work.

0.115 (epoch 10) and plateaus near 0.100 by epoch 18; the final cosine-decay phase contributes a further 0.003 improvement, consistent with fine-grained weight adjustment at low learning rates. The inverse correlation between AbsRel and $\delta < 1.25$ throughout training is shown in Figure 3 (top-right).

Loss decomposition. Figure 3 shows the per-step training dynamics over 11,160 gradient steps. SILog loss (Figure 3, middle-left) converges rapidly to 0.03–0.04 within the first 500 steps, indicating the model acquires correct metric scale early. The Sobel gradient loss (Figure 3, bottom) stabilises around 2.0 — an order of magnitude larger in absolute value but balanced by the 0.5 weighting coefficient. The two objectives do not conflict: SILog governs global scale accuracy while the gradient term enforces depth discontinuities at object boundaries.

Learning rate schedule. The cosine annealing schedule (Figure 3, middle-right) decays from 2×10^{-5} to approximately 2×10^{-7} by epoch 30. The steep final-phase decay enables fine-grained weight adjustment that accounts for the additional 0.003 AbsRel improvement between epoch 18 and the best epoch 27.

Transient instabilities. The spike in AbsRel and RMSE around epoch 5 coincides with the period of highest cosine-scheduled learning rate; the model recovers within one to two epochs. A similar RMSE excursion at epoch 14 is consistent with known sensitivity of metric depth to scale outliers at intermediate

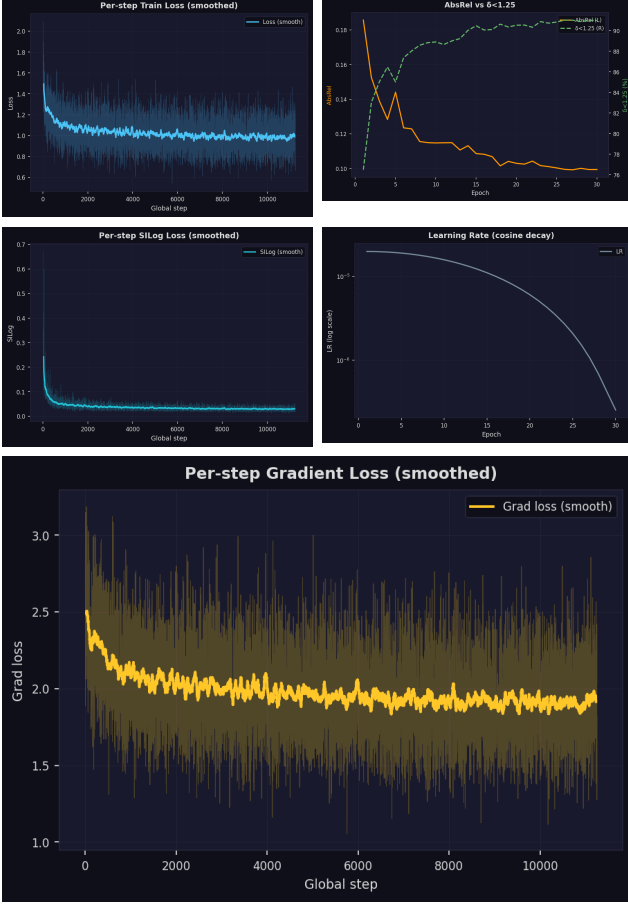


Figure 3. Per-step and supplementary training curves. **(top-left)** Per-step total train loss (smoothed); **(top-right)** AbsRel vs. $\delta < 1.25$ dual-axis overlay (epoch-level); **(middle-left)** Per-step SILog loss (smoothed); **(middle-right)** Learning rate on log scale (cosine annealing); **(bottom)** Per-step Sobel gradient loss (smoothed).

step sizes. A linear warm-up phase could mitigate these transients in future runs.

Limitations. The model was trained exclusively on MIMIR-UW synthetic data. Sim-to-real performance on actual AUV footage depends on residual domain gap not captured by MIMIR-UW’s optical simulation. The pretrained DA3 baseline was not quantitatively evaluated on the underwater validation set, preventing a direct improvement measurement; this evaluation is deferred to future work. Ablation of individual preprocessing components and the gradient loss is similarly deferred due to HPC walltime constraints.

7. Conclusion

This work presents a parameter-efficient domain adaptation pipeline for underwater monocular metric depth estimation. By combining rank-8 LoRA fine-tuning of

the DINOv2-L attention layers in Depth Anything 3 with physics-aware preprocessing — gray world white balance and percentile histogram stretching — we adapt a state-of-the-art terrestrial depth foundation model to the synthetic underwater domain using only 1% of trainable parameters. Training on the MIMIR-UW synthetic corpus for 30 epochs on a single A100 GPU achieves AbsRel = 0.099, RMSE = 0.739 m, and $\delta < 1.25 = 91.0\%$ on the held-out validation set, substantially surpassing the 0.15–0.20 initial target. The combined SILog and Sobel gradient objective provides stable, scale-correct convergence without conflicting gradient signals.

Future work includes: (1) quantitative evaluation of the pretrained DA3 baseline and ablation variants to isolate component contributions; (2) validation on real underwater AUV footage to characterise the sim-to-real transfer gap; (3) extension to multi-view DA3 for rigs with multiple synchronised cameras; and (4) knowledge distillation of the LoRA-adapted model for edge deployment on embedded AUV hardware.

References

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